



Robobarista makes latte

Cornell University's PR2 is now fully capable of making a latte using a set of instructions from the Internet <http://dgit.in/Robobarista>



Singing and MRI

Researchers have mapped the muscles that are involved during singing <http://dgit.in/MRISinging>

our bodily fluids and any damage to this essential organ will lead to extremely expensive medical procedures like Hemodialysis, which is administered to patients suffering from various stages of kidney failure. Administering hemodialysis is a very expensive affair and for an average Indian, most of the times it is not affordable. Of all the instruments and consumables involved in dialysis, the cumulative expense is the highest for a hemodialyzer. It accounts for 58% of the dialysis costs incurred per session of dialysis.

The instruments are not manufactured in India and all hemodialyzers are imported from Germany, Korea or Japan. The challenge in manufacturing a

microns and thickness of 35 microns. The estimated manufacturing cost of such an indigenous dialyzer is expected to be around `200 - 300. High-cut off (HCO) membranes for treatment of myeloma patients are being developed as well. One such HCO membrane costs `50,000-70,000, but using this technology, it is expected to come down to `500. This is a project of national importance and was funded by the Department of Science and Technology, Govt. of India, and industrial partners of the technology are FORUS Health Pvt. Ltd and Renalix, both companies based in Bangalore.

Dr. Shyam Vasudeva Rao and Dr. Lloyd Vincent (Senior Consultant Neph-

rologist, Narayana Hrudayalaya, and Bangalore) are the Directors of the two companies respectively and they are looking into the commercialization of the product. According to Dr. Lloyd Vincent, "The fibers match the clinical specifications of the market leaders and already reported biocompatible

polymers have been used to minimize the chances of adverse effects during dialysis". Animal and clinical trials of the product are awaited, although in-vitro tests have been completed.

The product won the Runners up prize at the "5th National Award for Technology Innovation in various Field of Petrochemicals and Downstream Plastic Processing Industry" which was presented by the Chemicals and Fertilisers minister, Shri Ananth Kumar at Bangalore on the 21st of February 2015.

This product, once commercialized, can equip India with the technology to not only sustain itself in the field of dialysis, but also project itself as a major exporter of such hemodialyzers to South-East Asian countries, Africa and South America. Efforts are also on to make

an end-to-end solution for dialysis by Renalix and IIT Kharagpur.

Solar-powered cold storage

IIT-KGP engineers have developed a solar-powered cold storage system that works at zero running cost as a solution to the wastage of agricultural produce in India. The facility has been developed at the Science and Technology Entrepreneurship Park (STEP) of IIT-Kharagpur funded by TIETS, Govt. Of India, under the company banner, Ecofrost Technologies, by mechanical engineering graduates - Devendra Gupta, Prateek Singhal & Vivek Pandey. In the National University Competition 'DuPont: The Power of Shunya', this innovation has won a first prize.

The micro cold storage system has been tested and proven itself in a farmland in Karnataka. The first of its kind in the world, there is no running costs for this unit and it works on sustainable technology throughout the year. A patent for the technology has been applied for, Pandey said. A target to manufacture 20,000 cold storage units in the next five years has been set up from the Pune unit.

Using solar panels and a uniquely designed thermal storage methodology that controls compartment cooling in tandem with regular cooling this micro cold storage increases the shelf life of agricultural produce. The power generated is sent to the compressor, running at various speeds to adjust to the cooling demand. Instead of batteries, there is a thermal storage unit which can store power for more than 36 hours for cloudy or rainy weather.

Every year India loses around 30 per cent of food production due to wastage and contamination according to studies. This technology can drastically cut down on that.

"The cost of the solution depends on the commodity to be stored. Normally, a fruit farmer should be able to recover the cost within three years and earn at least twice of what he is currently earning," said Gupta. The team plans to sell unit directly to farmers and to create village-level entrepreneurs who will act as nodal points for cold storage in markets where a farmer can store his produce at a fixed cost. Website: <http://www.ecofrost.in>



hemodialyzer is in spinning the clinical grade hollow fiber membranes. India does not possess the technology to spin hollow fiber membranes of an inner diameter 180-220 microns and a thickness of 15-40 microns. Only 4-5 companies worldwide enjoy the monopoly in this business and all have their own patented technology. Due to this, each dialyzer costs `1000 - 1500 and ideally three such dialyzers are needed per week for each patient.

The present innovation is about a technology that has been designed to spin such clinical grade fibers in India. This technology does not use the conventional and expensive spinnerets which are employed by the companies abroad. Instead, simple, disposable syringe assemblies have been used to spin dialysis grade fibers having an internal diameter of 220